

NMR LipoProfile® With Insulin Resistance Markers Without Lipids (Without Graph)

Test Details

METHODOLOGY

Nuclear magnetic resonance (NMR)

RESULT TURNAROUND TIME

2 - 3 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Historical reporting (LDL-P); insulin resistance & diabetes risk markers (large VLDL-P, small LDL-P, large HDL-P, VLDL size, LDL size, HDL size); insulin-resistance score; lipoprotein particle number (LDL-P); particle concentration and size (total HDL-P, small LDL-P, LDL size).

Use

NMR LDL-P is a management tool used in appropriate high-risk patients (type 2 diabetes mellitus, metabolic syndrome, CVD risk equivalent, statin-treated patients) to adjudicate response to treatment and guide adjustment in therapy. It is used in conjunction with other lipid measurements and clinical evaluation to aid in the management of lipoprotein disorders associated with cardiovascular disease.

Limitations

This test was developed and its performance characteristics determined by Labcorp. It has not been cleared or approved by the Food and Drug Administration.

Custom Additional Information

The NMR LipoProfile® test is a blood test that directly measures the amount of LDL circulating in the body. “LDL” is low-density lipoprotein and has long been recognized as a major causal factor in the development of heart disease. Although the relationship of increased LDL particle number and plaque buildup in the artery wall has been known since the 1950s, a diagnostic test did not exist to measure LDL particle number (LDL-P). Historically, LDL cholesterol, or LDL-C, has been used to estimate LDL levels to assess a patient’s LDL-related cardiovascular risk and judge an individual’s response to LDL-lowering therapy. The NMR LipoProfile® test reports results for LDL-P, a more reliable measure of LDL that directly counts the number of LDL particles a patient has using NMR technology.

Specimen Requirements

Specimen

Spun NMR LipoTube (preferred), serum from a plain red-top tube, plasma from a lavender-top (EDTA-no gel), **or** green-top (heparin-no gel) tube. Keep refrigerated and **ship with frozen cool packs.**

Volume

2 mL

Minimum Volume

1 mL

Container

NMR LipoTube (black-and-yellow-top tube) is the preferred container; plain red-top tube, lavender-top (EDTA-no gel) tube **or** green-top (heparin-no gel) tube.

Collection Instructions

Collect specimen in NMR LipoTube (black-and-yellow-top tube), which is the preferred container. Plain red-top, green-top (heparin-no gel) or lavender-top (EDTA-no gel) tubes also are acceptable. Serum or plasma drawn in gel-barrier collection tubes other than the NMR LipoTube should **not** be used. **The LipoTube is the only acceptable gel-barrier tube.**

Gently invert tube eight to 10 times to mix contents and allow specimen to clot for 30 minutes upright at room temperature prior to centrifugation (plasma tubes should not clot). Centrifuge specimen within two hours of collection at 1600 to 1800 xg for 10 to 15 minutes to separate serum/plasma from the red cells and to avoid red cell contamination during shipment. If the sample cannot be centrifuged immediately, the sample should be refrigerated (at 2°C to 8°C) and centrifuged within 24 hours of collection. **Note:** Centrifuging the specimen while still cold may negatively affect the migration of the gel to the serum/red cell interface and may increase the likelihood of specimens being contaminated with red cells during shipment. All specimens should be centrifuged by the client **prior** to shipment to Labcorp to ensure sample integrity. Do **not** open NMR LipoTube (black-and-yellow-top). Immediately after centrifugation, pipette separated red-top serum or green-top/lavender-top plasma into a transport tube and label accordingly (serum, heparin plasma, EDTA plasma). Keep samples refrigerated until shipment to the laboratory, and **ship with frozen cool packs.**

Storage Instructions

Refrigerate all acceptable tube types as soon as possible after centrifugation and within 24 hours of collection. Keep refrigerated prior to shipment, and **ship on frozen cool packs.** Do **not** store at room temperature. Do **not** freeze the sample. Sample is stable refrigerated for six days.

Patient Preparation

Patient should be fasting for eight hours.

Causes for Rejection

Unspun specimens; plasma/serum contaminated with red cells; citrated plasma (light blue-top tube); gross hemolysis; specimen received in inappropriate container; specimen stored at room temperature for more than a total preanalytical time of 24 hours; specimen more than six days old
